1.3 Solving One-Step Equations

Name _____ Period ____ Date _____

Algebra I 1.3 · Student sheet

I can…

- solve equations using addition or subtraction.
- solve equations using multiplication or division.
- write and solve equations for real-life problems.

Vocabulary · fill in the missing word

Equation — a statement that two _____ are equal

Solution — a value that makes the _____ true

Inverse operations — operations that _____ each other (+ and −, × and ÷)

Properties of equality — doing the same thing to both sides keeps an _____ true

Part 1 · Undo it

Equation	What was done to the variable?	Undo with	Solution	Check
$x - 7 = 12$				
$y + 5 = -2$				
$4n = -28$				
$\frac{m}{3} = 6$				
$k - 1.5 = -2.5$				
$(\frac{2}{3})p = 10$				

Part 2 · Notes and examples (fill in each note, then work the example)

Inverse operations: Inverse operations _____ each other: addition and _____; multiplication and _____.

Properties of equality: Whatever you do to one side, do to the _____.

Example 1

Solve each equation.

a. $x - 6 = 11$ b. $w + 3.5 = 1$

Example 2

Solve each equation.

a. $-7y = 42$ b. $\frac{t}{4} = -3$

Fractions: To undo multiplying by a fraction, multiply by its _____.

Example 3

Solve $(\frac{3}{4})x = 9$.

Example 4

You buy 4 movie tickets for $46. Write and solve an equation to find the cost of one ticket.

Part 3 · Partner practice

Partner 1

Solve $n + 12 = 5$.

Partner 2

Solve $-\frac{x}{5} = 2$.